PYL122 Assignment 1
Tobias Toll

Instructions: *Solve the problems during the tutorial session. Hand in the solutions to the TA one day before the following tutorial. Submit hardcopy to MS207c-19, or soft copy to gaurab.Kumar.Dash@physics.iitd.ac.in. Assignment will be considered for marks if handed in on time.*

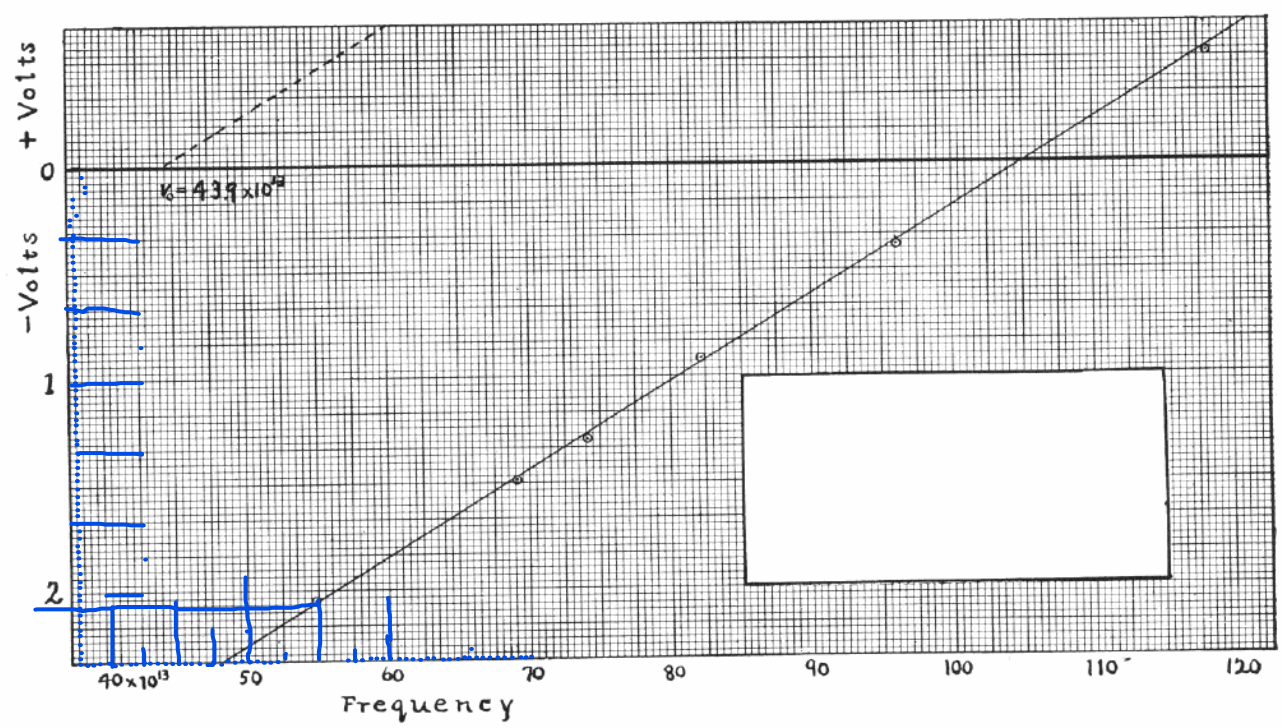
1. Given Planck's formula for the spectrum of a blackbody's radiation: $\omega(\nu, T) = \frac{8\pi}{c^2} \frac{h\nu^3}{e^{\frac{h\nu}{kT}} - 1}$,

(a) Calculate Wien's constant σ in Wien's diplacement law: $\lambda_{\max} = \frac{\sigma}{T}$.

Hints: One solution is that you use $\omega(\nu, T) = \left|\frac{\mathrm{d}I}{\mathrm{d}\nu}\right|$, and first do a variable transform $\nu \to \lambda$, with $\lambda\nu = c$ (don't forget the Jacobian). Then differentiate the resulting $\omega(\lambda, T) = \left|\frac{dI}{\mathrm{d}\lambda}\right|$ with respect to λ. The result is not analytically solvable, but one can estimate the solution fairly easy by trial and error.

(b) Show that the Black Body radiation spectrum does not overlap for different temperatures.

2. This is the data of the photo-electric effect collected by Millikan:



The horisontal axis shows the frequency of the light shining in $10^{13}\mathrm{s}^{-1}$, and the vertical axis shows the minimum voltage necessary to stop the current induced by the light. Use it to calculate Planck's constant.

3. (a) The Compton wavelength is defined as $\lambda_c = \frac{h}{mc}$. At what angle is light scattered, when the change in wavelength is one unit of the Compton wavelength?

(b) Light of 1nm scatter with an electron at rest. What momentum does the electron obtain?